

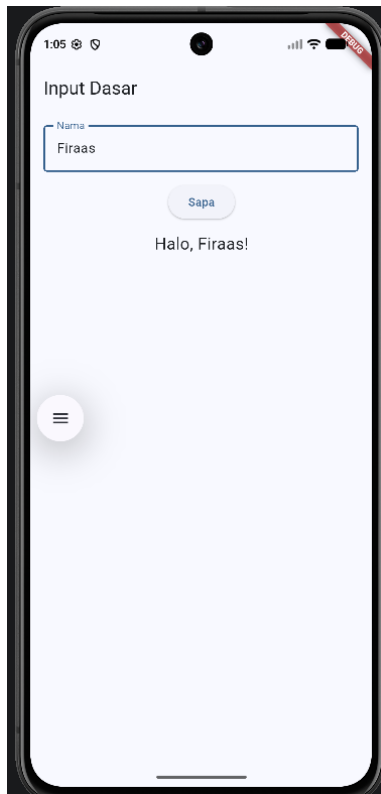
LANGKAH PRAKTIKUM BAGIAN A

Mengganti seluruh main.dart menjadi:

```
1 import 'package:flutter/material.dart';
2 void main() => runApp(const MyApp());
3 class MyApp extends StatelessWidget {
4   const MyApp({super.key});
5   @override
6   Widget build(BuildContext context) {
7     return MaterialApp(
8       title: 'Praktikum 3',
9       theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
10      home: const InputPage(),
11    );
12  }
13 }
14 class InputPage extends StatefulWidget {
15   const InputPage({super.key});
16   @override
17   State<InputPage> createState() => _InputPageState();
18 }
19 class _InputPageState extends State<InputPage> {
20   final _controller = TextEditingController();
21   String _salam = '';
22   @override
23   void dispose() {
24     _controller.dispose();
25     super.dispose();
26   }
27 }
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Input Dasar')),
    body: Padding(
      padding: const EdgeInsets.all(16),
      child: Column(
        children: [
          TextField(
            controller: _controller,
            decoration: const InputDecoration(
              labelText: 'Nama',
              border: OutlineInputBorder(),
            ),
          ),
          const SizedBox(height: 12),
          ElevatedButton(
            onPressed: () {
              setState(() => _salam = 'Halo, ${_controller.text}!');
            },
            child: const Text('Sapa'),
          ),
          const SizedBox(height: 12),
          Text(_salam, style: const TextStyle(fontSize: 20)),
        ],
      ),
    ),
  );
}
```

HASIL:



Aplikasi dapat diinput pada text box, ketika mengklik tompol sapa maka akan mengeluarkan output sapaan “Halo, Firaas!”.

LANGKAH PRAKTIKUM BAGIAN B:

Buat halaman baru FormPage (di file yang sama), lalu ubah home: menjadi const FormPage().

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 3',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const InputPage(),
    );
  }
}

class InputPage extends StatefulWidget {
  const InputPage({super.key});

  @override
  State<InputPage> createState() => _InputPageState();
}

class _InputPageState extends State<InputPage> {
  final _controller = TextEditingController();
  String _salam = "";

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Input Dasar'),
```

```

actions: [
  IconButton(
    icon: const Icon(Icons.assignment),
    tooltip: 'Ke Form Pendaftaran',
    onPressed: () {
      Navigator.push(
        context,
        MaterialPageRoute(builder: (context) => const FormPage()),
      );
    },
  ),
],
),
body: Padding(
  padding: const EdgeInsets.all(16),
  child: Column(
    children: [
      TextField(
        controller: _controller,
        decoration: const InputDecoration(
          labelText: 'Nama',
          border: OutlineInputBorder(),
        ),
      ),
      const SizedBox(height: 12),
      ElevatedButton(
        onPressed: () {
          setState(() => _salam = 'Halo, ${_controller.text}!');
        },
        child: const Text('Sapa'),
      ),
      const SizedBox(height: 12),
      Text(_salam, style: const TextStyle(fontSize: 20)),
    ],
  ),
),
);
}
}

```

```

class FormPage extends StatefulWidget {
  const FormPage({super.key});
}

```

```

@override
State<FormPage> createState() => _FormPageState();
}

class _FormPageState extends State<FormPage> {
  final _formKey = GlobalKey<FormState>();
  final _nama = TextEditingController();
  final _email = TextEditingController();
  String? _jurusan;
  bool _setuju = false;

  @override
  void dispose() {
    _nama.dispose();
    _email.dispose();
    super.dispose();
  }

  void _ kirim() {
    if (_formKey.currentState!.validate()) {
      final jurusan = _jurusan ?? '-';
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(content: Text('Terdaftar: ${_nama.text} ($jurusan)'),
      );
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Form Pendaftaran')),
      body: Form(
        key: _formKey,
        child: ListView(
          padding: const EdgeInsets.all(16),
          children: [
            TextFormField(
              controller: _nama,
              decoration: const InputDecoration(
                labelText: 'Nama lengkap',
                border: OutlineInputBorder(),
              ),
              validator: (v) =>

```

```

        (v == null || v.trim().isEmpty) ? 'Nama wajib diisi' : null,
    ),
    const SizedBox(height: 12),
    TextFormField(
      controller: _email,
      keyboardType: TextInputType.emailAddress,
      decoration: const InputDecoration(
        labelText: 'Email',
        border: OutlineInputBorder(),
      ),
      validator: (v) {
        if (v == null || !v.contains('@')) return 'Email tidak valid';
        return null;
      },
    ),
    const SizedBox(height: 12),
    DropdownButtonFormField<String>(
      decoration: const InputDecoration(
        labelText: 'Jurusan',
        border: OutlineInputBorder(),
      ),
      items: const [
        DropdownMenuItem(value: 'TI', child: Text('Teknik Informatika')),
        DropdownMenuItem(value: 'SI', child: Text('Sistem Informasi')),
        DropdownMenuItem(value: 'TE', child: Text('Teknik Elektro')),
      ],
      onChanged: (v) => setState(() => _jurusan = v),
      validator: (v) => v == null ? 'Pilih jurusan' : null,
    ),
    CheckboxListTile(
      title: const Text('Saya menyetujui ketentuan'),
      value: _setuju,
      controlAffinity: ListTileControlAffinity.leading,
      onChanged: (v) => setState(() => _setuju = v ?? false),
    ),
    const SizedBox(height: 12),
    ElevatedButton(
      onPressed: _setuju ? _ kirim : null,
      child: const Text('Daftar'),
    ),
  ],
),
),
),

```

```
);  
}  
}
```

Sesuai arahan modul saya mengubah home: const InputPage(), dengan home: const FormPage(), agar aplikasi langsung menampilkan kelas halaman FormPage saat dijalankan.

LANGKAH PRAKTIKUM BAGIAN C

Pada Bagian A dan B, state hidup di dalam satu halaman (setState). Bayangkan aplikasi daftar tugas: halaman daftar menampilkan tugas, sedangkan halaman tambah membuat tugas baru. Kedua halaman perlu berbagi data yang sama. Mengoper data lewat constructor dan callback dari satu halaman ke halaman lain cepat menjadi rumit. Solusinya: pindahkan data ke satu objek bersama yang dapat diakses kedua halaman. Inilah peran Provider.

LANGKAH PRAKTIKUM BAGIAN D

Langkah 1

Pasang paket pada terminal di folder proyek: flutter pub add provider

Langkah 2

```
class Tugas {  
  String judul;  
  bool selesai;  
  Tugas(this.judul, {this.selesai = false});  
}  
  
class TugasModel extends ChangeNotifier {  
  final List<Tugas> _items = [];  
  List<Tugas> get items => List.unmodifiable(_items);  
  int get jumlahSelesai => _items.where((t) => t.selesai).length;  
  void tambah(String judul) {  
    _items.add(Tugas(judul));  
    notifyListeners();  
  }  
}
```

```

}

void toggle(int index) {
  _items[index].selesai = !_items[index].selesai;
  notifyListeners();
}

void hapus(int index) {
  _items.removeAt(index);
  notifyListeners();
}
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => TugasModel(),
      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Tugas',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const TugasPage(),
    );
  }
}

```

Langkah 3

```
class TugasPage extends StatelessWidget {  
  const TugasPage({super.key});  
  
  @override  
  Widget build(BuildContext context) {  
    final model = context.watch<TugasModel>();  
    return Scaffold(  
      appBar: AppBar(  
        title: Text('Tugas (${model.jumlahSelesai}/${model.items.length}')),  
      ),  
      body: ListView.builder(  
        itemCount: model.items.length,  
        itemBuilder: (context, i) {  
          final t = model.items[i];  
          return ListTile(  
            leading: Checkbox(  
              value: t.selesai,  
              onChanged: (_) => context.read<TugasModel>().toggle(i),  
            ),  
            title: Text(  
              t.judul,  
              style: TextStyle(  
                decoration: t.selesai ? TextDecoration.lineThrough : null,  
              ),  
            ),  
            trailing: IconButton(  
              icon: const Icon(Icons.delete),  
              onPressed: () => context.read<TugasModel>().hapus(i),  
            ),  
          );  
        },  
      ),  
    );  
  }  
}
```

```

);
},
),
floatingActionButton: FloatingActionButton(
  onPressed: () {
    Navigator.push(
      context,
      MaterialPageRoute(builder: (_) => const TambahPage()),
    );
  },
  child: const Icon(Icons.add),
),
);
}
}

```

Langkah 4

```

class TambahPage extends StatefulWidget {
  const TambahPage({super.key});

  @override
  State<TambahPage> createState() => _TambahPageState();
}

class _TambahPageState extends State<TambahPage> {
  final _controller = TextEditingController();

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }
}

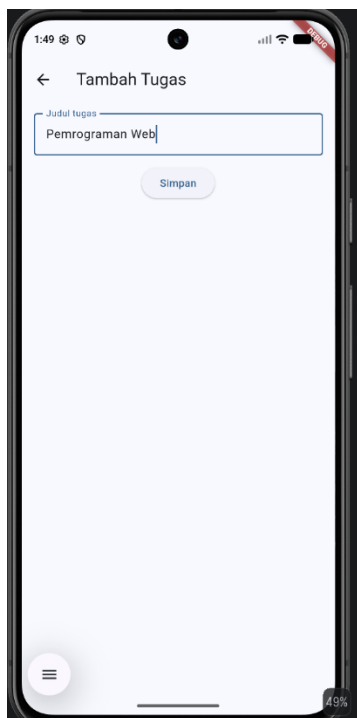
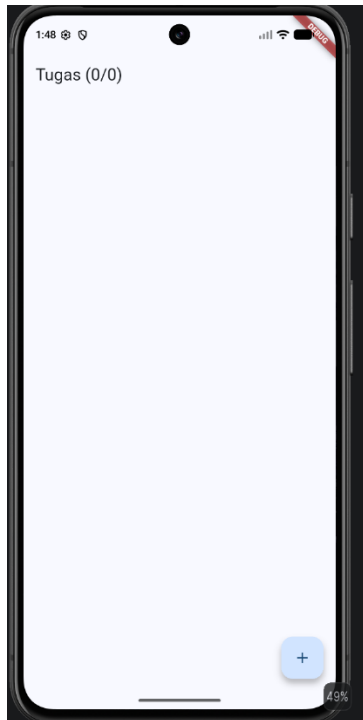
```

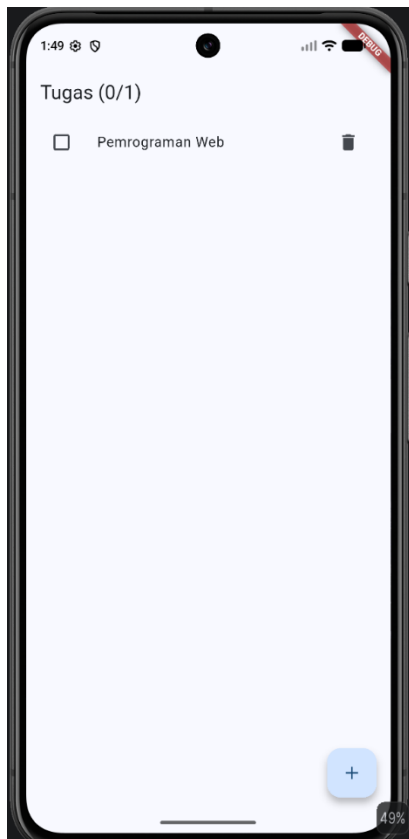
```
void _simpan() {  
  final judul = _controller.text.trim();  
  if (judul.isEmpty) return;  
  context.read<TugasModel>().tambah(judul);  
  Navigator.pop(context);  
}  
  
@override  
Widget build(BuildContext context) {  
  return Scaffold(  
    appBar: AppBar(title: const Text('Tambah Tugas')),  
    body: Padding(  
      padding: const EdgeInsets.all(16),  
      child: Column(  
        children: [  
          TextField(  
            controller: _controller,  
            autofocus: true,  
            decoration: const InputDecoration(  
              labelText: 'Judul tugas',  
              border: OutlineInputBorder(),  
            ),  
            onSubmitted: (_) => _simpan(),  
          ),  
          const SizedBox(height: 12),  
          ElevatedButton(onPressed: _simpan, child: const Text('Simpan')),  
        ],  
      ),  
    ),  
  );  
}
```

}

}

HASIL:





TUGAS MANDIRI

1. Tambahkan validasi pada TambahPage: judul minimal 3 karakter dan tampilkan pesan error (ubah ke TextFormField + Form).
2. Tambahkan method hapusSelesai() pada TugasModel dan tombol di AppBar untuk menghapus semua tugas yang sudah selesai.
3. Tampilkan SnackBar "Tugas ditambahkan" setelah tugas disimpan.
4. Bila daftar kosong, tampilkan teks "Belum ada tugas" di tengah layar.

```
import 'package:flutter/material.dart';  
import 'package:provider/provider.dart';
```

```
class Tugas {  
  String judul;  
  bool selesai;  
  
  Tugas(this.judul, {this.selesai = false});  
}
```

```
class TugasModel extends ChangeNotifier {
```

```

final List<Tugas> _items = [];

List<Tugas> get items => List.unmodifiable(_items);

int get jumlahSelesai => _items.where((t) => t.selesai).length;

void tambah(String judul) {
  _items.add(Tugas(judul));
  notifyListeners();
}

void toggle(int index) {
  _items[index].selesai = !_items[index].selesai;
  notifyListeners();
}

void hapus(int index) {
  _items.removeAt(index);
  notifyListeners();
}

void hapusSelesai() {
  _items.removeWhere((t) => t.selesai);
  notifyListeners();
}

}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => TugasModel(),
      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Tugas',

```

```

    theme: ThemeData(
      colorSchemeSeed: Colors.blue,
      useMaterial3: true,
    ),
    home: const TugasPage(),
  );
}
}

```

```

class TugasPage extends StatelessWidget {
  const TugasPage({super.key});

```

```

  @override

```

```

  Widget build(BuildContext context) {
    final model = context.watch<TugasModel>();

```

```

    return Scaffold(
      appBar: AppBar(
        title: Text('Tugas (${model.jumlahSelesai}/${model.items.length}')),
        actions: [
          if (model.jumlahSelesai > 0)
            IconButton(
              icon: const Icon(Icons.delete_sweep),
              tooltip: 'Hapus Tugas Selesai',
              onPressed: () {
                context.read<TugasModel>().hapusSelesai();
              },
            ),
        ],
      ),

```

```

      body: model.items.isEmpty
        ? const Center(
          child: Text(
            'Belum ada tugas',
            style: TextStyle(fontSize: 16, color: Colors.grey),
          ),
        )
        : ListView.builder(
          itemCount: model.items.length,
          itemBuilder: (context, i) {
            final t = model.items[i];
            return ListTile(

```

```

        leading: Checkbox(
          value: t.selesai,
          onChanged: (_) => context.read<TugasModel>().toggle(i),
        ),
        title: Text(
          t.judul,
          style: TextStyle(
            decoration:
              t.selesai ? TextDecoration.lineThrough : null,
          ),
        ),
        trailing: IconButton(
          icon: const Icon(Icons.delete),
          onPressed: () => context.read<TugasModel>().hapus(i),
        ),
      );
    },
  ),
  floatingActionButton: FloatingActionButton(
    onPressed: () {
      Navigator.push(
        context,
        MaterialPageRoute(builder: (_) => const TambahPage()),
      );
    },
    child: const Icon(Icons.add),
  ),
);
}
}

```

```

class TambahPage extends StatefulWidget {
  const TambahPage({super.key});

  @override
  State<TambahPage> createState() => _TambahPageState();
}

```

```

class _TambahPageState extends State<TambahPage> {
  final _formKey = GlobalKey<FormState>();
  final _controller = TextEditingController();

  @override

```

```

void dispose() {
  _controller.dispose();
  super.dispose();
}

void _simpan() {
  if (_formKey.currentState!.validate()) {
    final judul = _controller.text.trim();

    context.read<TugasModel>().tambah(judul);
    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(
        content: Text('Tugas ditambahkan'),
        duration: Duration(seconds: 2),
      ),
    );

    Navigator.pop(context);
  }
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Tambah Tugas')),
    body: Padding(
      padding: const EdgeInsets.all(16),

      child: Form(
        key: _formKey,
        child: Column(
          children: [
            TextFormField(
              controller: _controller,
              autofocus: true,
              decoration: const InputDecoration(
                labelText: 'Judul tugas',
                border: OutlineInputBorder(),
              ),
            ),
            validator: (value) {
              if (value == null || value.trim().isEmpty) {
                return 'Judul tugas tidak boleh kosong';
              }
            }
          ]
        )
      )
    )
  );
}

```

```
        if (value.trim().length < 3) {
            return 'Judul minimal 3 karakter';
        }
        return null;
    },
    onFieldSubmitted: (_) => _simpan(),
),
const SizedBox(height: 12),
ElevatedButton(
    onPressed: _simpan,
    child: const Text('Simpan'),
),
],
),
),
),
);
}
```